USER MANUAL

AI Academic Mentor

Personalized Learning & Study Assistant

Installation · Login · Features · System Usage

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
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Section	7C
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> **NOTE** This manual describes the AI Academic Mentor web application exactly as built in the project source code and UI design documents. Screenshots are taken from the running application.

1. Introduction

1.1 Purpose of this manual

This manual teaches students and administrators how to install, sign in to, and use **AI Academic Mentor**. It is written for readers with no programming background for the “using” chapters (3, 4 and 5); Chapter 2 (Installation) is aimed at whoever sets the system up, such as a teammate, lab technician or instructor.

1.2 What is AI Academic Mentor?

AI Academic Mentor is a web-based study platform for university students. It uses generative AI (Google Gemini, with Hugging Face as a backup provider) to answer academic questions, build quizzes, summarize lecture notes and create personalized study plans. It also tracks learning progress and gives administrators a dashboard to manage users.

1.3 Who uses it?

User type	What they can do
Student	Register, sign in, chat with the AI, generate and take quizzes, summarize notes, create study plans, view progress, read notifications, manage personal data.
Administrator	Everything a student can do, plus the Admin Panel: view platform statistics, search and filter users, promote / demote / disable / enable / delete accounts, export the user list.

1.4 Feature summary

Feature	Description
Dashboard	Greeting, live statistics, quick actions and recent activity.
AI Chatbot	Multi-conversation academic Q&A assistant.
Quiz Generator	AI-written MCQ, True/False or mixed quizzes with scoring and explanations.
Note Summarizer	Turns pasted notes into concise, detailed, bullet-point or exam-ready summaries.
Study Planner	AI-generated 3, 5 or 7-day plans with a session checklist.
Progress Analytics	Score trends, topic performance, weekly activity and quiz history.
Notifications	In-app log of important events with read / clear controls.
Settings	Profile information, data management and “About” information.

Feature	Description
Admin Panel	User management and platform data for administrators.

1.5 System requirements

Item	Requirement
Operating system	Windows 10/11, macOS, or Linux
Node.js	Version 18 or newer (the backend declares `node >= 18`)
npm	Installed together with Node.js
Database	A free MongoDB Atlas cluster (cloud) and its connection string
AI key	A Google Gemini API key (must start with `AIza`); a free Hugging Face token is optional (backup)
Browser	Current Google Chrome, Microsoft Edge, Firefox or Safari
Internet	Required (database and AI services are online)
Screen	Desktop (1440 px) and mobile (420 px) layouts are both supported

1.6 Technology overview

The system has two independent parts that run separately: a **frontend** (React 19 + Vite, shown in the browser) and a **backend** (Node.js + Express API with MongoDB). The frontend sends requests to the backend, which checks the user's identity (JWT token), talks to the AI providers and stores data in MongoDB.

2. Installation

Follow these steps in order. The project contains two folders, `backend` and `frontend`. They are separate npm projects, so each is installed and started on its own.

2.1 Before you begin

1. Install **Node.js 18 or newer** from nodejs.org (the LTS version is recommended). Verify in a terminal: `node -v` and `npm -v`.
2. Create a free **MongoDB Atlas** account at cloud.mongodb.com, create a cluster, add a database user, and allow your IP address under Network Access. Copy the connection string.
3. Create a **Gemini API key** at aistudio.google.com/app/apikey. (Optional: create a Hugging Face token at huggingface.co/settings/tokens for the backup AI provider.)
4. Unzip the project so you have a folder named `AI-Academic-Mentor` containing `backend`, `frontend` and `README.md`.

> **IMPORTANT** Never share or publish your `.env` file. It contains your database password, JWT secret and API keys. Only the template file `.env.example` should ever be shared.

2.2 Install and configure the backend

Step 1: Install dependencies

```
cd AI-Academic-Mentor/backend
npm install
```

Step 2: Create the environment file

Copy the template to a real `.env` file and edit it:

```
cp .env.example .env        # on Windows: copy .env.example .env
```

Open `.env` in a text editor and fill in the values described below.

Variable	Meaning / what to enter
`PORT`	Port the API listens on. Default `5000`.
`NODE_ENV`	`development` while testing, `production` when deployed.
`MONGODB_URI`	Your MongoDB Atlas connection string (replace `<username>` and `<password>`).
`JWT_SECRET`	A long random secret used to sign login tokens. Generate one with: `node -e "console.log(require('crypto').randomBytes(64).toString('hex'))"`

Variable	Meaning / what to enter
`JWT_EXPIRES_IN`	How long a login lasts. Default `7d` (7 days).
`GEMINI_API_KEY`	Your Gemini key; it must start with `AIza`.
`HF_ACCESS_TOKEN`	Optional Hugging Face token used when Gemini fails.
`ADMIN_ACCESS_CODE`	Secret code required to register an administrator account. Change it from `CHANGE_ME`.
`CLIENT_URL`	The address of the frontend, for CORS. For local use set `http://localhost:5173` (see warning below).

> **IMPORTANT** The template defaults `CLIENT_URL` to `http://localhost:3000`, but the frontend development server runs on **port 5173**. Set `CLIENT_URL=http://localhost:5173` in `.env`, otherwise the browser will block every request with a "CORS blocked" error.

Step 3: (Optional) Create demo accounts

The seed script creates sample administrator and student accounts in your database:

```
npm run seed
```

It is safe to run more than once; existing accounts are skipped. The accounts are listed in Section 3.5.

Step 4: Start the backend

```
npm start            # production-style start
npm run dev          # auto-restart on file changes (nodemon)
```

When it starts correctly, the terminal shows a banner with the port, followed by **"MongoDB connected successfully"** and a line confirming the Gemini key was loaded.

2.3 Install and start the frontend

Open a **second** terminal window (keep the backend running) and run:

```
cd AI-Academic-Mentor/frontend
npm install
npm run dev
```

Vite prints a local address, normally **http://localhost:5173**. Open it in your browser to see the AI Academic Mentor home page.

The frontend finds the backend automatically at `http://localhost:5000` when you browse from `localhost`. If your backend runs elsewhere, create a file `frontend/.env` containing:

```
VITE_API_BASE=http://localhost:5000
```

2.4 Verify the installation

1. Open **http://localhost:5000/api/health** in a browser. You should see JSON containing `"status": "ok"` and `"db": "connected"`.
2. Open **http://localhost:5173**. The landing page with the heading “Study Like Never Before With AI” should appear.
3. Click **Sign In** and log in with a demo account (Section 3.5). You should land on the Dashboard.

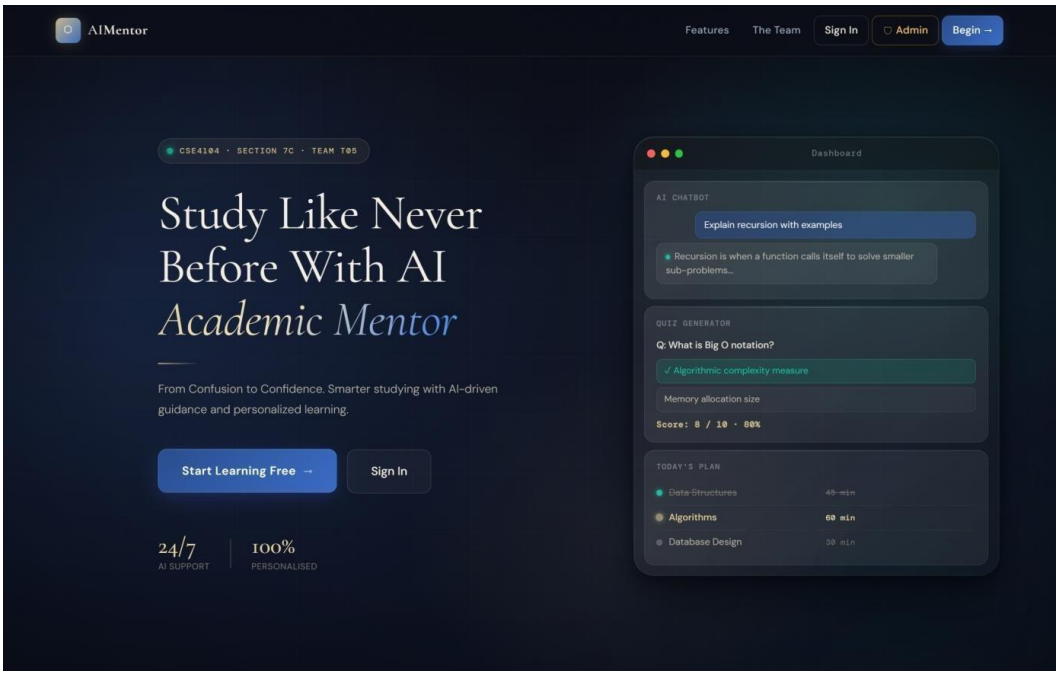


Figure 1: Landing page shown after a successful installation

2.5 Deploying online (optional)

For a public deployment, host the backend on a service such as Render and the frontend on Netlify. Before deploying: (1) set `VITE_API_BASE` (or `PRODUCTION_API_BASE` in `frontend/src/services/api.js`) to your real backend URL; (2) set `CLIENT_URL` on the backend to your Netlify address (any `*.netlify.app` address is already accepted); (3) add all `.env` values in the hosting dashboard rather than uploading the file.

2.6 Installation troubleshooting

Problem	Cause and fix
`npm install` fails	Check that Node.js is version 18+ (`node -v`) and that you are inside the correct folder (`backend` or `frontend`).
“MongoDB connection error” and the server exits	Wrong `MONGODB_URI`, wrong password, or your IP is not allowed in Atlas Network Access. Correct it and restart.

Problem	Cause and fix
Browser shows “CORS blocked for origin”	`CLIENT_URL` does not match the address in your browser. Set it to `http://localhost:5173` and restart the backend.
Port 5000 already in use	Change `PORT` in `.env` and set `VITE_API_BASE` in `frontend/.env` to match.
AI answers say “static fallback”	Gemini and Hugging Face both failed. Check `GEMINI_API_KEY` (must start with `AIza`) and the backend log.
Page loads but nothing saves	The backend may not be running or is in demo mode (no database). Check the backend terminal.

3. Login Process

Every feature except the home page requires an account. Your session is kept in the browser, so you stay signed in until you choose **Sign Out** (or your token expires after 7 days).

3.1 Creating an account (Register)

1. On the home page click **Start Learning Free**, or go to `/register`.
2. Fill in the form: **Full Name** (at least 3 characters), **Email Address**, **Student ID** (optional, for example 1123012XXXX), **Department** (CSE, EEE, BBA, English or Other), **Password** (at least 6 characters) and **Confirm Password**.
3. Click **Create Account**.
4. On success you are signed in automatically, a welcome notification is added, and you are taken to the Dashboard.

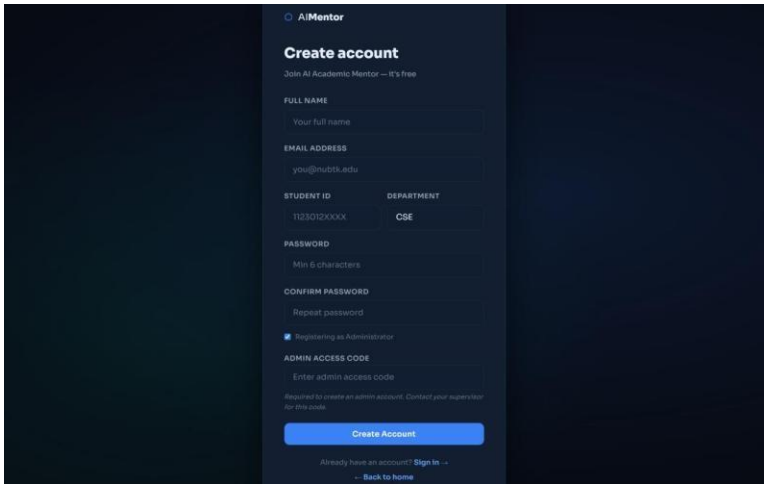


Figure 2: Create account page

Message	What it means
Passwords do not match.	The two password boxes differ. Retype them.
Password must be at least 6 characters.	Choose a longer password.
Enter the admin access code…	You ticked “Registering as Administrator” but left the code empty.
Registration failed. Is the backend server running?	The backend is offline or the email is already registered.

NOTE Ticking **Registering as Administrator** reveals an **Admin Access Code** box. Only people who know the server’s `ADMIN_ACCESS_CODE` can create an administrator account. Students should leave it unticked.

3.2 Signing in as a student

1. Open the home page and click **Sign In** (or go to `/login`).
2. Enter your **Email Address** and **Password**.
3. Click **Sign In**. The button shows “Signing in…” while the server checks your details.
4. You are taken to the **Dashboard**.

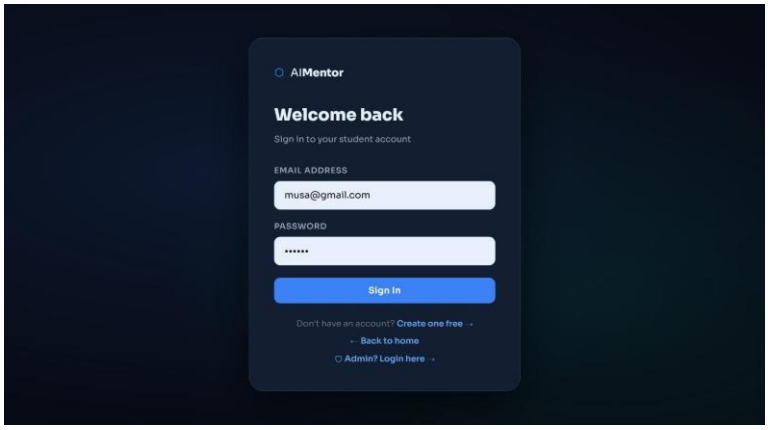


Figure 3: Sign-in page

3.3 Signing in as an administrator

1. On the sign-in page click **Admin? Login here** → (or open `/login?role=admin`). The card turns amber and shows an “Admin Login” badge.
2. Enter your administrator email and password and click **Sign In**.
3. You are taken to the **Admin Panel**. A student account used here is rejected with “This login is for administrators only.”

3.4 Signing out

Click your name and avatar at the bottom of the left sidebar (it shows “Sign out”), or use **Settings** → **Profile** → **Sign Out**. You return to the home page and must sign in again to continue.

3.5 Demo accounts

If you ran `npm run seed` (Section 2.2), these accounts exist:

Role	Email	Password
Administrator	`admin@nubtk.edu`	`admin123`
Student	`student@nubtk.edu`	`student123`
Other students	`alice@nubtk.edu`, `bob@nubtk.edu`, `chayon@nubtk.edu`	`Student@123`

IMPORTANT Demo passwords are public. Change or delete these accounts before using the system with real students.

3.6 Login error messages

Message	Meaning and action
Invalid email or password	Check spelling and Caps Lock, or register a new account.
Your account has been suspended. Contact your administrator.	An administrator disabled your account. Ask them to enable it.
This login is for administrators only…	You used the admin login with a student account. Use the normal Sign In.
Signed in, but the backend is NOT connected to MongoDB (demo mode)	The server has no database. Your data will not be saved; tell the system owner.
Too many requests, please try again later	Rate limit reached (100 requests per 15 minutes). Wait and retry.

4. Features

4.1 Navigation

After signing in, the **left sidebar** is your main menu. On phones it opens with the menu button at the top.

Sidebar item	Opens
Dashboard	Overview and quick actions
AI Chatbot	Ask academic questions
Quiz Generator	Create and take quizzes
Note Summarizer	Summarize study notes
Study Planner	Create study plans
Progress	Analytics and history
Notifications	Event log (a red badge shows the unread count)
Settings	Profile and data management
Admin Panel	Visible to administrators only

4.2 Dashboard

The Dashboard greets you by first name (“Good morning / afternoon / evening”) and summarizes your study activity.

- **Statistic cards:** Quizzes Taken, AI Conversations, Study Plans and Average Quiz Score. Values are refreshed from the database when the page loads.
- **Quick Actions:** one-click shortcuts to AI Chatbot, Quiz Generator, Summarizer and Study Planner.
- **Recent Activity:** your latest quizzes, questions and plans with relative times (for example “2h ago”).

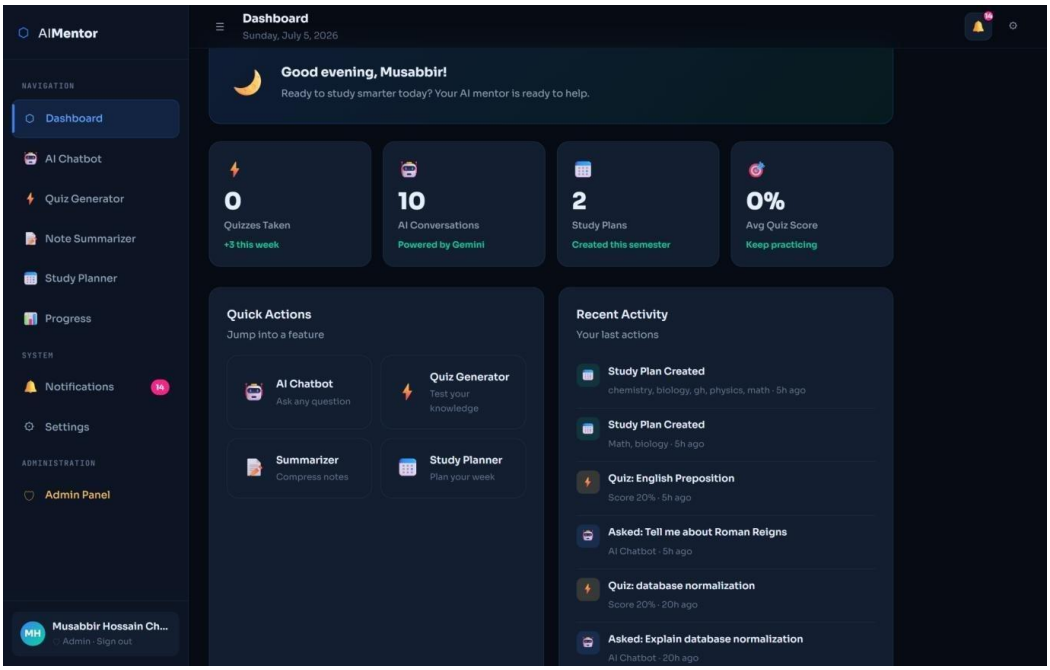


Figure 4: Dashboard

4.3 AI Chatbot

Ask any academic question and get an AI answer. Conversations are saved so you can return to them.

- **+ New Chat** starts a new conversation; earlier ones are listed on the left and named after your first question.
- Click a suggestion chip (for example “Explain recursion with examples”) or type your own question at the bottom.
- Press **Enter** to send; press **Shift + Enter** for a new line.
- **Clear** (top right) deletes the current conversation after confirmation. ← **Dashboard** returns to the home screen.

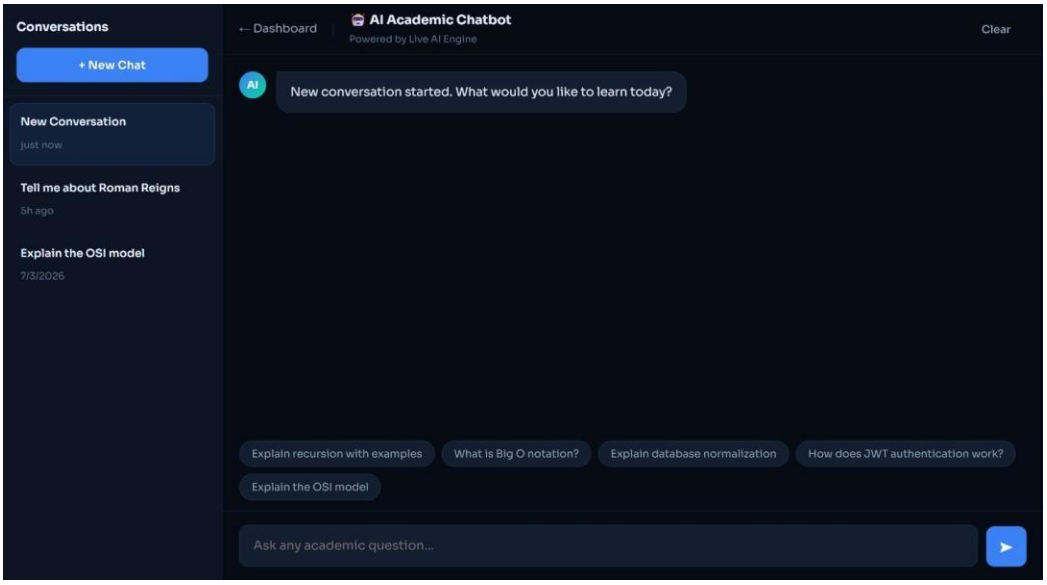


Figure 5: AI Chatbot with conversation list and suggestions

4.4 Quiz Generator

Creates a quiz on any topic. Use the left panel to choose the options:

Option	Choices
Topic / Subject	Free text, for example “Data Structures” or “OOP”
Number of Questions	3, 5 or 10
Difficulty	Easy, Medium or Hard
Quiz Type	Multiple Choice (MCQ), True / False, or Mixed

After) **Generate Quiz**, questions appear on the right. Select one answer per question (a progress bar shows how many are answered). When all are answered, click **Submit Quiz** →. You then see your score, a color for each answer, and a **Explanation** under every question. Scores of 70% or more are rated great, 50–69% okay, below 50% needs practice. Your result is saved to your history, to Progress and to the database. Use **New Quiz** to start again or **View Progress** to see your trend.

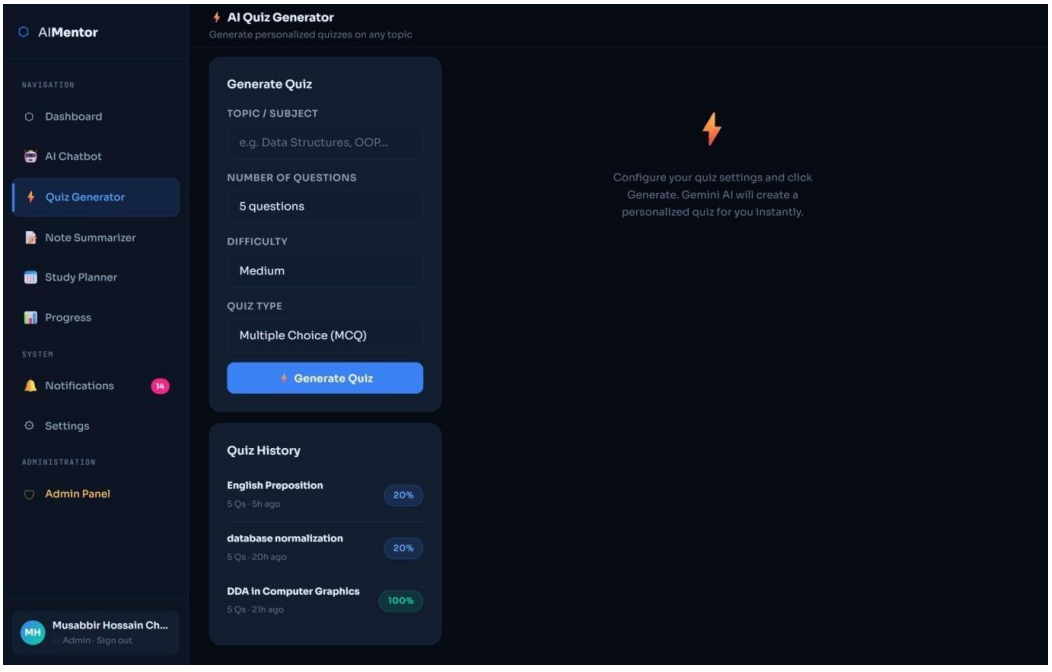


Figure 6: Quiz Generator settings and quiz history

4.5 Note Summarizer

Paste long notes and receive a clear summary in your chosen style.

Style	Best for
Concise	A short, memorable overview
Detailed	Full coverage with definitions and structure

Style	Best for
• Bullet Points	Hierarchical headings and sub-points
🎯 Exam Ready	Key concepts, definitions, likely exam topics and quick review points

Enter an optional **Subject / Topic**, paste text into **Your Notes** (at least 50 characters; a live character counter is shown) and click **Summarize Notes**. Use **Copy** to copy the result. Earlier summaries appear under **Previous Summaries**; click one to reopen it.

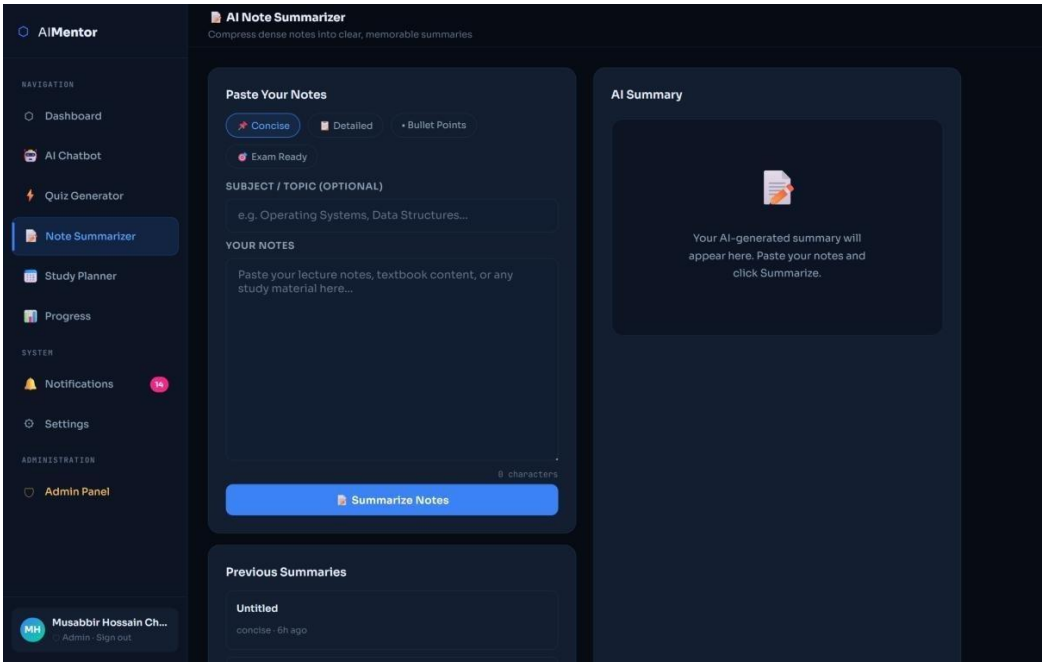


Figure 7: Note Summarizer

4.6 Study Planner

Builds a day-by-day schedule based on your subjects and available time.

Setting	Choices
Add Subject	Type a subject and press Enter or **+**; remove it with **×**. At least one is required.
Study Days	3, 5 or 7 days
Daily Hours	2, 3, 4 or 6 hours
Goal	Exam Prep, Revision, Learn New Topics, Assignments
Level	Beginner, Intermediate, Advanced
Additional Notes	Optional extra instructions, for example weak topics

Click **Generate Study Plan**. Each day lists sessions with subject, topic, duration, a type (lecture, practice, review or assignment), a priority and a study tip. Tick the checkbox on a session when you finish it; a completion bar tracks your percentage. Past plans are kept under **Saved Plans**.

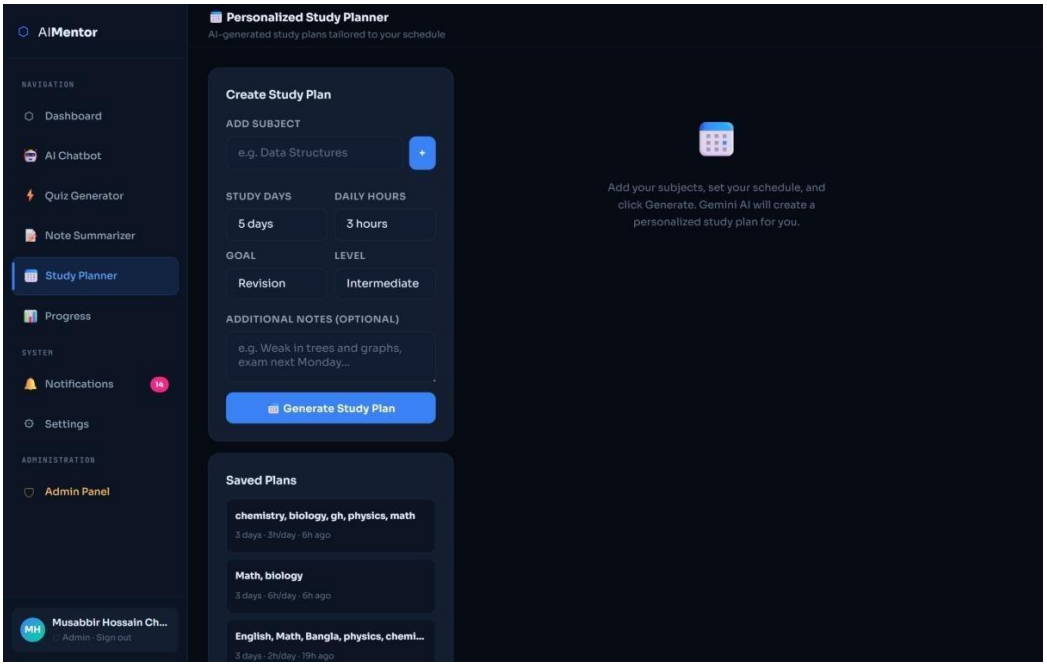


Figure 8: Study Planner form and saved plans

4.7 Progress Analytics

Shows how you are doing over time:

- **Cards:** Total Quizzes, Average Score, Best Score and Day Streak.
- **Charts:** Quiz Scores Over Time, Topic Performance, AI Feature Usage and Weekly Activity.
- **Quiz History table:** Topic, Score, Questions, Correct, Date and Grade.
- **Reset Data** (top right) clears all your progress after confirmation. This cannot be undone.

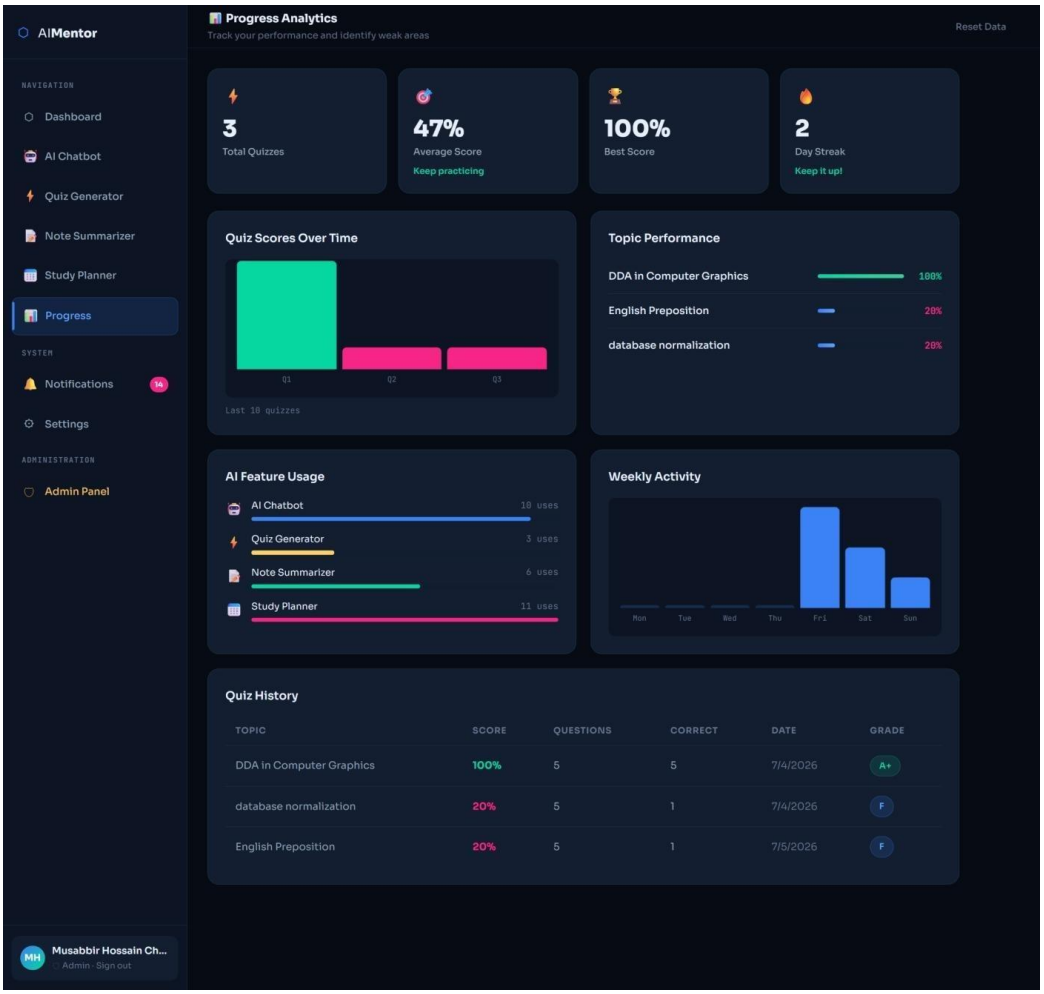


Figure 9: Progress Analytics

4.8 Notifications

The system records events such as signing in, finishing a quiz, creating a summary or a study plan. Unread items have a blue dot. Use **Mark all read** to clear the badge or **Clear all** to delete every notification (confirmation required).

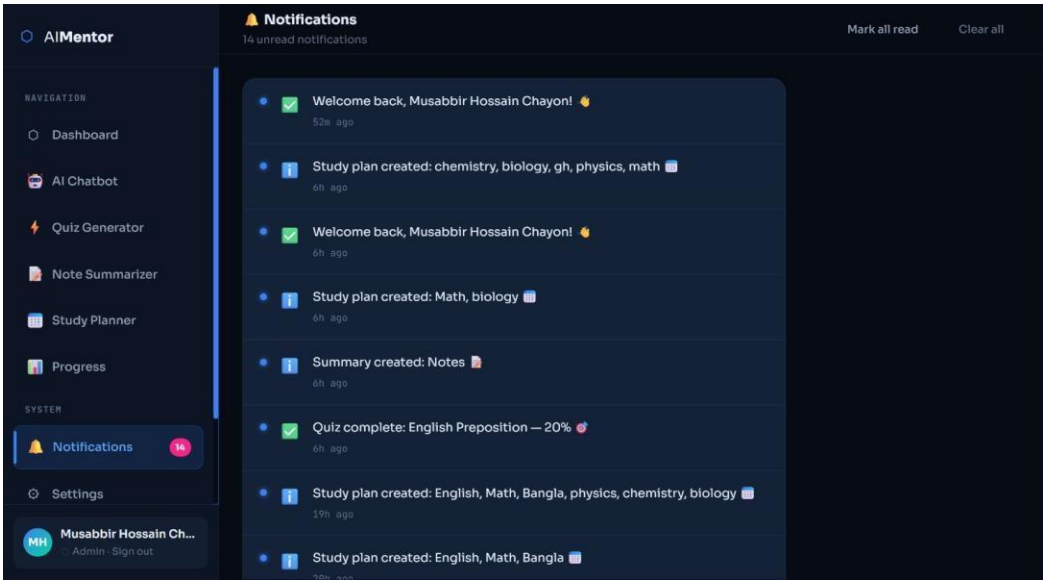


Figure 10: Notifications page

4.9 Settings

Settings has three tabs:

- **Profile:** shows your Name, Email, Student ID, Department, Role and Joined date, and the **Sign Out** button.
- **Data Management:** separate **Clear** buttons for quiz results, chat history, study plans and summaries, plus **Clear All Data**. Each asks for confirmation.
- **About:** version information, the AI engine (Google Gemini 2.5 Flash / Hugging Face) and project details.

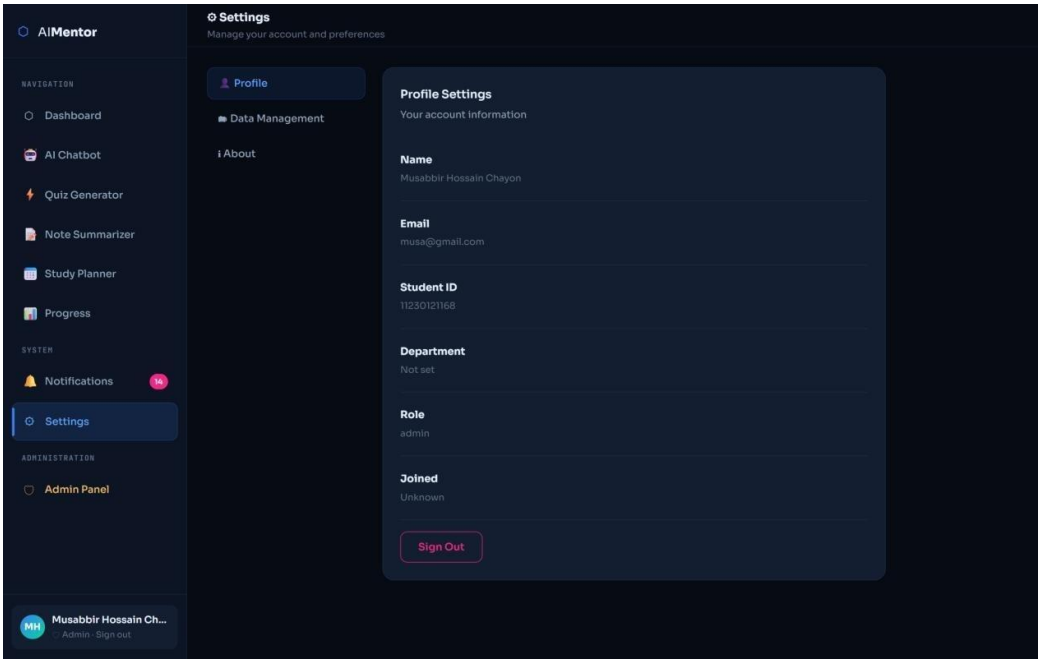


Figure 11: Settings, Profile tab

4.10 Admin Panel (administrators only)

The Admin Panel has four tabs and a statistics row (Total Users, Registered Accounts, Total Quizzes, Platform Avg Score).

Tab	What you can do
+ User Management	Search by name or email; filter All / Students / Admins / Banned; **View** a user's activity (quizzes, average score, chats, plans); **Make Admin** or **Demote**; **Disable** or **Enable** a login; **Delete** a user permanently together with their data. You cannot delete your own account.
Platform Data	Per-user data overview and **Export User List (JSON)**. Bulk Clear is intentionally disabled to protect real data.
Announcements	Form for title, message and priority. Note: this tab is a preview only in this version and does not yet broadcast to users.
⚙ System	System information (total and disabled accounts) and **Clear All Notifications**.

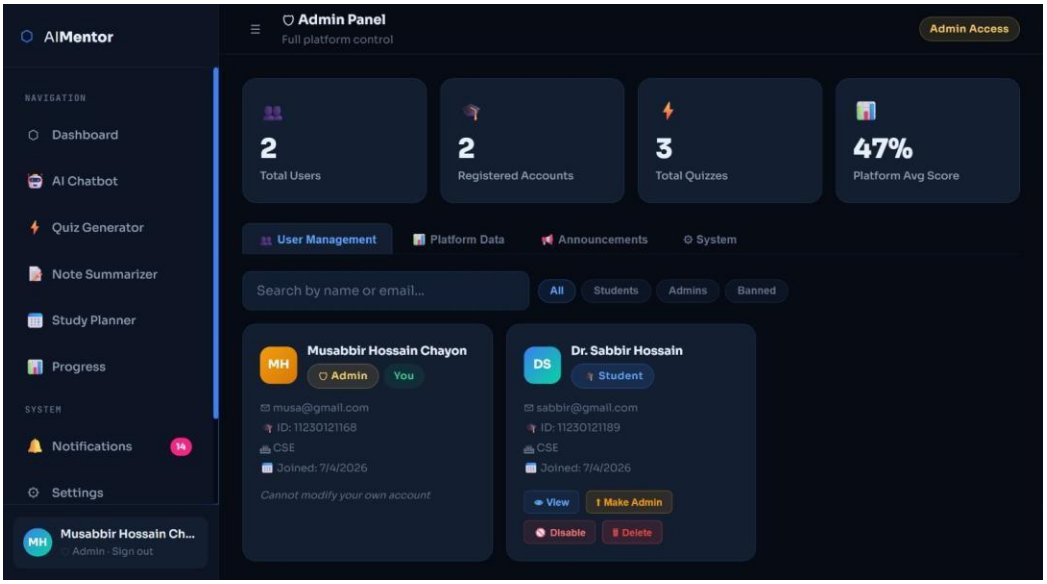


Figure 12: Admin Panel, User Management tab

5. System Usage

This chapter walks through complete tasks from start to finish.

5.1 Quick start for a new student (5 minutes)

1. Register an account (Section 3.1) and arrive at the Dashboard.
2. Open **AI Chatbot** and click a suggestion such as “What is Big O notation?” to see an AI answer.
3. Open **Quiz Generator**, enter a topic, keep the defaults and generate a 5-question quiz. Answer and submit.
4. Open **Progress** to see your first score appear in the charts.

5.2 Workflow: preparing for an exam

1. Go to **Study Planner**. Add your subjects, choose **Study Days** (for example 5), **Daily Hours** (for example 3) and the goal **Exam Prep**. Click **Generate Study Plan**.
2. Each study day, open the plan and tick sessions as you complete them.
3. Paste your lecture notes into **Note Summarizer**, choose **Exam Ready** and click **Summarize Notes**. Copy the result into your own notes.
4. Test yourself in **Quiz Generator** on the same topics at **Hard** difficulty.
5. Ask the **AI Chatbot** to explain any question you got wrong; the quiz explanations tell you which topics to ask about.
6. Check **Progress** → **Topic Performance** to find weak topics, and repeat.

5.3 Workflow: understanding a difficult concept

1. Open **AI Chatbot** and click **+ New Chat**.
2. Ask your question, for example “Explain database normalization with an example.”
3. Ask follow-up questions in the same conversation; the AI remembers the earlier messages.
4. Use **Quiz Generator** with the same topic to check your understanding.

5.4 Workflow: administrator tasks

Disable a user

1. Sign in through the admin login and open **Admin Panel** → **User Management**.
2. Find the user with the search box or filters.
3. Click **Disable** and confirm. The user can no longer sign in; click **Enable** to restore access.

Promote or demote a user

1. Locate the user and click **Make Admin** (or **Demote**).
2. Confirm the prompt. Admin accounts have full platform control, so promote carefully.

Review a user's activity

1. Click **View** on a user card.
2. A panel shows their quiz count, average score, chats and plans, plus account status.

Export the user list

1. Open the **Platform Data** tab.
2. Click **Export User List (JSON)** to download the list.

5.5 Using the system on a phone

The layout adapts to narrow screens. The sidebar becomes a slide-in menu opened from the top bar, cards stack vertically and the chatbot uses the full screen. All features work the same way as on the desktop.

5.6 How your data is stored

Data	Where it lives
Account, quiz results, study plans, summaries	MongoDB database (permanent, tied to your account)
Chat conversations, notifications, recent-activity lists	Your browser's local storage (only on this device and browser)
Sign-in session	Your browser, valid for 7 days or until you sign out

TIP Because chats and notifications are stored in the browser, clearing your browser data or switching devices removes them. Quiz scores, plans and summaries are kept on the server and are reloaded when you sign in.

5.7 Good practices

- Be specific in questions and topics; "binary search tree deletion" gives better results than "trees".
- Wait for the AI to finish before sending another message or generating another quiz.
- AI output can contain mistakes. Check important facts against your textbook or instructor.
- Use a strong, unique password and always sign out on shared computers.

6. Troubleshooting & FAQ

6.1 Common problems

Symptom	Solution
“Could not reach the backend” in the chatbot	Start the backend (`npm start` in `backend`), and confirm `VITE_API_BASE` points to it.
Quiz shows “Failed to generate quiz”	The AI returned an unusable answer or the AI key is invalid. Click Generate again; if it persists, ask the system owner to check the Gemini key.
AI answer ends with “static fallback”	Both AI providers failed. Check API keys and network; wait a minute if the free quota was reached.
“AI rate limit exceeded”	More than 60 AI requests in one minute. Wait a moment and retry.
I am sent back to the login page	Your session expired or you signed out. Sign in again.
I cannot sign in and see “suspended”	An administrator disabled your account. Contact them.
Summarizer says notes are too short	Enter at least 50 characters of notes.
Study Planner says “Add at least one subject”	Type a subject and press Enter or the **+** button first.
My chats disappeared	Chats are stored in the browser; they are lost if browser data is cleared or you use another device.

6.2 Frequently asked questions

Is the system free?

The registration is free. AI usage depends on the free quotas of the Gemini and Hugging Face accounts configured by the system owner.

Can I change my password or profile details?

The current Settings page displays your profile but does not provide a password-change form. Ask an administrator for help if you need changes.

Can other students see my data?

No. Quizzes, plans and summaries are linked to your account. Only administrators can view user activity summaries.

How do I delete my data?

Use **Settings** → **Data Management** to clear quizzes, chats, plans and summaries, or **Progress** → **Reset Data**.

Which AI is used?

Google Gemini (2.5 Flash) is primary; Hugging Face Inference is used as a backup if Gemini fails.

Appendix A. Reference

A.1 Application pages

Address	Page	Access
`/`	Landing page	Everyone
`/login`, `/register`	Sign in, Create account	Everyone
`/dashboard`	Dashboard	Signed-in users
`/chatbot`	AI Chatbot	Signed-in users
`/quiz`	Quiz Generator	Signed-in users
`/summarizer`	Note Summarizer	Signed-in users
`/planner`	Study Planner	Signed-in users
`/progress`	Progress Analytics	Signed-in users
`/notifications`	Notifications	Signed-in users
`/settings`	Settings	Signed-in users
`/admin`	Admin Panel	Administrators only

A.2 Backend API endpoints

Used by the application internally; useful for administrators and testers (for example with Postman).

Group	Endpoints
Auth	`POST /api/auth/register`, `POST /api/auth/login`, `GET /api/auth/me`, `PUT /api/auth/profile`, `POST /api/auth/logout`
AI	`POST /api/ai/chat`, `/api/ai/quiz`, `/api/ai/summarize`, `/api/ai/plan`
Quiz	`GET/POST /api/quiz`, `GET/PUT/DELETE /api/quiz/:id`
Study plans	`GET/POST /api/plan`, `GET/PUT/DELETE /api/plan/:id`
Summaries	`GET/POST /api/summary`, `GET/PUT/DELETE /api/summary/:id`
Notes	`GET/POST /api/notes`, `GET/PUT/DELETE /api/notes/:id`
Progress	`GET /api/progress`, `GET /api/progress/stats`
Admin	`GET /api/admin/stats`, `GET /api/admin/users`, `GET/PUT/DELETE /api/admin/users/:id`, `GET /api/admin/users/:id/activity`

Group	Endpoints
Health	`GET /api/health`

A.3 Glossary

Term	Meaning
API	The set of server addresses the website calls to get and save data.
JWT	JSON Web Token: a signed “ticket” proving you are signed in.
MongoDB Atlas	The cloud database where accounts, quizzes and plans are stored.
Gemini	Google’s generative AI model used for answers, quizzes, summaries and plans.
CORS	A browser safety rule; the backend must list the frontend’s address as allowed.
Demo mode	A state where the backend runs without a database, so nothing is saved permanently.

A.4 Document information

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